

Taylor series

In calculus, a Taylor series is an infinite sum of terms that represents a function as a power series around a specific point. It is one of the most powerful tools in analysis and approximations. Given a function $f(x)$ that is infinitely differentiable around a point a , its Taylor series expansion around a is given by:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(a)}{n!} (x-a)^n$$

where:

- $f^{(n)}(a)$ is the n -th derivative of $f(x)$ evaluated at a ,
- $n!$ is the factorial of n , and
- $(x-a)^n$ represents powers of $(x-a)$.

If a is set at 0, it is also called the Maclaurin series.

Taylor Series for Common Functions

1. Exponential Function

The Taylor series expansion of e^x around $a = 0$ (also known as the Maclaurin series) is:

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$$

2. Sine Function

The Taylor series for $\sin(x)$ around $a = 0$ is:

$$\sin(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n+1}}{(2n+1)!} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$$

3. Cosine Function

The Taylor series for $\cos(x)$ around $a = 0$ is:

$$\cos(x) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{2n}}{(2n)!} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$$

Remainder Term

In practice, we truncate the sum after a finite number of terms. The Taylor series only approximates the function up to a certain order. The error introduced by truncating the series is given by the remainder term, which is:

$$R_n(x) = \sum_{j=n+1}^{\infty} \frac{f^{(j)}(a)}{j!} (x-a)^j = \frac{f^{(n+1)}(c)}{(n+1)!} (x-a)^{n+1}$$

for some c between a and x .

The Taylor series is a fundamental tool in calculus for approximating functions near a given point. In statistics, it is widely applied for methodological developments, numerical implementations, and theoretical explorations.

If $|x - a| < \epsilon$ (small) i.e. we are interested in learning the function locally around a , then a Taylor series-based approximation for f with first n terms only would have a maximum error of $\frac{f^{(n+1)}(c)}{(n+1)!} \epsilon^{n+1}$. Thus, with more terms, the error decreases exponentially $[\epsilon^{n+1} = \exp\left(-(n+1)\frac{1}{\log \epsilon}\right)]$. Oftentimes, setting $n = 2$ is sufficient to get a good approximation. In the case of multivariate functions (i.e. functions with multiple inputs), the derivatives beyond the second are hard to work with as they form multidimensional arrays. The matrix based on the second-order partial derivative of a multivariate function is also called the Hessian matrix. Second-order Taylor expansion is the max we go for multivariate functions, and thus it is applied assuming $\mathbf{x}$ and $\mathbf{a}$ are close, ensuring a smaller approximation error.

For $x \in (a - h/2, a + h/2)$ with small h , we can make the approximation $f(x) \approx f(a) + (x - a)f^{(1)}(a)$ with absolute error $|R_1(x)| \leq \frac{f^{(2)}(c)}{2!} (h)^2$.

In the case of second order approximation near a , $f(x) \approx f(a) + (x-a)f^{(1)}(a) + (x-a)^2 f^{(2)}(a)$ with absolute error $|R_2(x)| \leq \frac{f^{(3)}(c)}{3!} (h)^3$. Thus, for a fixed $h < 1$, higher-order polynomials allow more accurate approximations.